Exchange rate and Production network

Macro Internal Seminar

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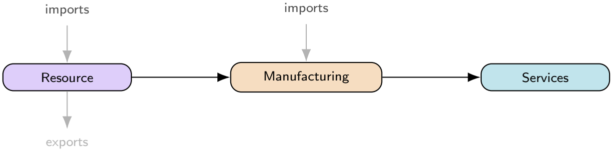
Outline

Motivation

Network Phillips curves (Rubbo 2024, La'O–Tahbaz-Salehi 2022): **closed economy**, unique DC index

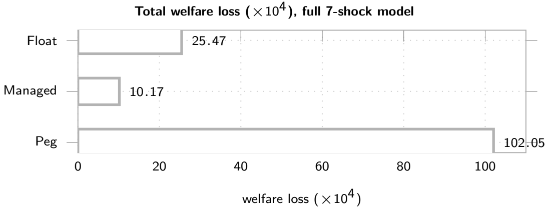
Open-economy NK (Galí–Monacelli 2005): FX channel, **no network**

This paper: both together — multi-sector network + FX



Questions: Optimal FX regime? Does the DC index survive a 2nd cost-push channel?

Headline Result



Managed float dominates both corners

Peg dominated: zero autonomy, UIP shock passes straight through

Driver: persistence (near-unit-root NFA), not volatility

Units: consumption-equivalent loss, $\times 10^{-4}$. Calibrated to Chile's real national input-output table (Banco Central de Chile, Cuadros 12x12, CdeR 2018, ref. year 2023) — full calibration and sources on the next two slides. Solved as the genuine nonlinear model (Dynare), not linearized.

Literature

Network side (closed economy):

Rubbo (2024) — Calvo network $\Rightarrow$ unique DC index (1 cost-push channel)
La'O & Tahbaz-Salehi (2022) — optimal MP in production networks
Pasten, Schoenle & Weber (2020) — sectoral rigidity heterogeneity drives MP transmission

Open-economy side:

Galí & Monacelli (2005) — 1-sector SOE: domestic-inflation targeting $\approx$ optimal, peg dominated (ToT channel)
Fanelli & Straub (2021) — FX intervention as a *second* instrument, separate from i_t
Schmitt-Grohé & Uribe (2003) — debt-elastic premium to stationarize NFA; flags 1st-order welfare risk
Gopinath & Itskhoki (2022); Amiti et al. (2019); Auer, Levchenko & Sauré (2019) — pass-through, dominant currency, IO spillovers
Itskhoki & Mukhin (2023); Bianchi & Coulibaly (2025); Coulibaly (2023) — frontier: managed float/fear-of-floating *optimal*, not behavioral (no network)
Gnocato, Montes-Galdón & Stamato (2025, ECB); Kalemli-Özcan et al. (2025) — open economy *with* network (tariffs), fixed Taylor rule, no FX-regime choice[1]
Qiu, Wang, Xu & Zanetti (2026, JME) — closest paper: derives the open-economy DC index directly extending Rubbo, WIOD 43-country calibration; but *static* (no NFA/persistence), no UIP, and no FX-regime choice — their own conclusion lists relaxing financial autarky as their top open extension

Gap: network + open economy + FX **regime choice** together — no one has this. Same ranking as Galí–Monacelli but a **different channel** (risk-premium, not ToT); Managed's dominance echoes Fanelli–Straub & Itskhoki–Mukhin, but from network persistence.

$\rightarrow$ paper-by-paper comparison to the literature: Appendix

[1] Details & what these papers cite for open-economy production networks: Appendix.

This Paper

Extend Rubbo $\rightarrow$ SOE: imported inputs Ω^F, UIP, debt-elastic premium

Ask: optimal FX regime? DC index survive with 2 cost-push channels?

Three literatures, none overlapping fully (network only; open economy only; network + open economy but no regime choice) — **first to put all three together**

Not just Galí-Monacelli with sectors: ranking survives, but the mechanism (risk-premium) and sector heterogeneity are new

Calibrate: Chile's real I-O table (Banco Central), also Korea, Czechia & a stylized baseline; solve **nonlinear** model directly

Model Components

Small open economy: home is a price- and rate-taker in world markets — foreign price P_t^{F*}, foreign rate i^*, and foreign demand D_t^* are exogenous; home cannot affect them.

Six building blocks, presented in order:

1. **Households** — consumption/labor choice, foreign bond holdings
2. **Firms: technology** — Cobb-Douglas production, marginal cost
3. **Production network** — Leontief inverse, Domar weights
4. **Foreign sector** — law of one price, UIP, net foreign assets
5. **Equilibrium** — goods, labor, asset market clearing
6. **Firms: price setting** — Calvo pricing, Phillips curve
7. **Monetary policy** — float / peg / managed float

Households

The representative household maximizes

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

subject to the budget constraint:

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1}) B_{t-1} + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*) \right] S_t B_{t-1}^*$$

B_t (domestic bond) nets to zero in equilibrium — identical households, no domestic counterparty.

The consumption aggregator (home bundle, Cobb-Douglas across sectors, vs. imports) is:

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \qquad C_t^H = \prod_i (C_{it}^H)^{\beta_i^H}$$

where:

C_t, L_t: cons., labor	β, γ, φ: disc., RRA, Frisch^{-1}	P_t^C: CPI	W_t: wage
B_t, B_t^*: dom./for. bond	i_{t-1}, i^*: dom./for. rate	S_t: exch. rate	Π_t: profits
$\psi, \bar{b}^*$: debt prem., target	C_t^H, C_t^F: home/for. goods	ω, η: bias, elast.	β_i^H: sector i share

Firms: Technology

A unit mass of firms $f \in [0,1]$ share the *same* Cobb-Douglas technology, each producing a differentiated variety:

$$Y_{ift} = A_{it}\, L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H}\; M_{ift}^{\Omega_i^F}$$

Varieties aggregate via a CES (Dixit-Stiglitz) index; firm f faces isoelastic demand:

$$Y_{it} = \Big(\int_0^1 Y_{ift}^{\frac{\varepsilon_i - 1}{\varepsilon_i}}\, df \Big)^{\frac{\varepsilon_i}{\varepsilon_i - 1}}, \qquad Y_{ift} = \Big(\frac{P_{ift}}{P_{it}} \Big)^{-\varepsilon_i} Y_{it}$$

CRS $\Rightarrow$ marginal cost is **common across firms** in sector i:

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H}\, (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Two cost-push channels: MC_{it} moves with A_{it} (TFP) *and* S_t (import cost). Absent frictions, firms would set $P_{ift} = \frac{\varepsilon_i}{\varepsilon_i - 1} MC_{it}$ (markup over cost); with **Calvo pricing**, only a fraction δ_i resets each period — mechanics & Phillips curve: later slide.

where:

Y_{ift}: firm f output	ε_i: sector elast.	P_{ift}: firm f price	P_{it}: sector price idx.
MC_{it}: marg. cost	W_t: wage	P_{jt}: sector j price	$S_t P_t^{F*}$: import price

Production Network

Stacking each firm's input shares Ω^H_{ij}, Ω^F_i across sectors gives the **input-output network** $\Omega^H \in \mathbb{R}^{N\times N}$ and import-exposure vector $\boldsymbol{\omega}_F \in \mathbb{R}^N$.

The **Leontief inverse** sums direct and indirect linkages:

$$(I-\Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \cdots$$

applied to consumption shares $\boldsymbol{\beta}^H$ (Households) it gives each sector's **Domar weight** (total GDP sales share); applied to $\boldsymbol{\omega}_F$, its total foreign exposure:

$$\boldsymbol{\lambda}^\top = (\boldsymbol{\beta}^H)^\top (I-\Omega^H)^{-1}, \qquad \mathcal{M} = (I-\Omega^H)^{-1}\boldsymbol{\omega}_F$$

where:
Ω^H_{ij}: dom. input share Ω^F_i: import share $\boldsymbol{\omega}_F = (\Omega^F_1, \ldots, \Omega^F_N)^\top$ $(I-\Omega^H)^{-1}$:
$\boldsymbol{\beta}^H$: cons. shares (Households) $\boldsymbol{\lambda}$: Domar weights $\mathcal{M}$: total (direct+indirect) foreign expo

Firms: Price Setting

Calvo pricing through the network delivers the (vector) Phillips curve:

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t \boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma \Delta e_t$$

Same amplifier, two shocks: $\mathcal{B}, \Gamma$ are the *same* rigidity-adjusted operator $\tilde{A} \equiv \Delta(I - \Omega^H \Delta)^{-1}$, applied to labor share $\boldsymbol{\alpha}$ vs. import share $\boldsymbol{\omega}_F$:

$$\mathcal{B} = \frac{\tilde{A}\boldsymbol{\alpha}}{\kappa}, \qquad \Gamma = \frac{\tilde{A}\boldsymbol{\omega}_F}{\kappa}$$

Only relative shocks matter: $\mathcal{V}\mathbf{1} = \mathbf{0}$ — uniform TFP shocks create no cost-push; only sector-specific gaps $\boldsymbol{\chi}_t = (I - \Omega^H)^{-1} \log \mathbf{A}_t - \boldsymbol{p}_{t-1}$ do.

where:

$\boldsymbol{\pi}_t, \tilde{y}_t$: infl., gap	$\boldsymbol{\chi}_t$: TFP cost-push	Δe_t: exch. rate chg.	κ: GE scalar
$\mathcal{B}, \Gamma$: PC slope, FX pass-thru	$\mathcal{V}$: TFP cost-push matrix	$\tilde{A}$: rigidity-adj. Leontief	

Foreign Sector

P_t^{F*}, i^*, D_t^* are exogenous AR(1) processes. Law of one price:

$$P_t^{FH} = S_t P_t^{F*}$$

UIP, with a debt-elastic risk premium closing the model (Schmitt-Grohé–Uribe) and RP_t the risk-premium shock studied below:

$$i_t = i^* + \mathbb{E}_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

NFA accumulates the trade balance; exports respond to relative prices:

$$\hat{b}_t^* = \frac{1+i_{t-1}}{\Pi_t^C}\hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}, \qquad EX_t = \kappa_{EX} D_t^* \left(\frac{P_t^H}{P_t^{FH}}\right)^{-\theta^*}$$

Two channels to the domestic economy:

$$\hat{s}_t + \hat{p}_t^{F*} \;\to\; MC_{it} \;\to\; \pi_{it} \quad \text{(cost-push)} \qquad \hat{s}_t \;\to\; EX_t \;\to\; \tilde{y}_t \;\to\; \pi_{it} \quad \text{(demand)}$$

where:

P_t^{FH}: import price	$\hat{b}_t^*$: NFA (scaled)	i_{t-1}: dom. rate	Π_t^C: CPI infl.
EX_t, IM_t: exports, imports	κ_{EX}: export scale	D_t^*: for. demand	θ^*: export elast.
$\hat{s}_t$: log exch. rate	$\hat{p}_t^{F*}$: log for. price	$\tilde{y}_t$: output gap	
i^*: foreign rate (fixed)	$\psi(\cdot)$: debt-elastic wedge	RP_t: risk-premium shock (AR(1))	

Monetary Policy Regimes

Three exchange-rate regimes are compared. All target the **DC index π_t^{DC}, not CPI inflation**: Rubbo (2024) shows CPI targeting does *not* achieve divine coincidence even in the closed economy — π_t^{DC} is the closed-economy-optimal benchmark, so it is the natural rule to test for survival once the economy is opened.

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

The exchange rate adjusts endogenously.

Hard peg

$$\hat{s}_t = 0$$

Adjustment occurs through domestic inflation and output.

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The central bank partially stabilizes the exchange rate.

$\hat{i}_t$: policy rate π_t^{DC}: DC index $\tilde{y}_t$: output gap $\hat{s}_t$: exchange rate

Equilibrium System

Everything above collapses to four blocks — this is what is actually coded up and simulated in Dynare.

Phillips curve (vector, network-amplified, two cost-push channels):

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

IS curve (aggregate demand; open economy adds the same relative-price term as the cost-push channel above):

$$\tilde{y}_t = \mathbb{E}_t\tilde{y}_{t+1} - \frac{1}{\gamma}\left(i_t - \mathbb{E}_t\pi_t^C - r_t^{\mathrm{nat}}\right) - \frac{\eta(1-\omega)}{\gamma}\mathbb{E}_t\Delta\left(\hat{s}_t + \hat{p}_t^{F*} - \hat{p}_t^H\right)$$

Foreign sector (UIP closes the FX block; NFA is the only state carried across regimes):

$$i_t = i^* + \mathbb{E}_t\Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t, \qquad \hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C}\hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}$$

Policy rule (regime-dependent; the only block that changes across Float/Peg/Managed — same rules as previous slide):

$$\hat{i}_t = \begin{cases} \phi_\pi\pi_t^{DC} + \phi_y\tilde{y}_t & \text{Float} \\ \text{residual } (\hat{s}_t = 0) & \text{Peg} \\ \phi_\pi\pi_t^{DC} + \phi_y\tilde{y}_t + \phi_s\hat{s}_t & \text{Managed} \end{cases}$$

i_t above is the level rate (Households, Foreign Sector); $\hat{i}_t$ is its log-deviation, the Taylor-rule instrument (Monetary Policy Regimes).

where:

Model Overview

Small open economy with sectoral production networks.

Foreign variables are exogenous:

$$P_t^{F*}, \qquad i^*, \qquad D_t^*$$

Model blocks

1. Households
2. Firms
3. Production network
4. Foreign sector
5. Price setting
6. Monetary policy
7. Equilibrium

Households

Representative household chooses consumption, labor, and foreign assets.

$$E_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right]$$

subject to

$$P_t^C C_t + B_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i_{t-1}) B_{t-1} + (1+i^*) \left[1 - \psi(\hat{b}_{t-1}^* - \bar{b}^*) \right] S_t B_{t-1}^*$$

Consumption basket

$$C_t = \left[\omega^{1/\eta} (C_t^H)^{\frac{\eta-1}{\eta}} + (1-\omega)^{1/\eta} (C_t^F)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}$$

Key features

- Domestic and imported goods are CES substitutes.
- Domestic goods are Cobb–Douglas across sectors.
- Foreign bond holdings determine net foreign assets.

Firms

Each sector produces differentiated varieties using labor, domestic intermediates, and imported inputs.

$$Y_{ift} = A_{it} L_{ift}^{\alpha_i} \prod_j X_{ijft}^{\Omega_{ij}^H} M_{ift}^{\Omega_i^F}$$

Marginal cost

$$MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

Implications

Higher wages increase costs.

Imported inputs create exchange-rate pass-through.

TFP and exchange rates generate two cost-push channels.

Production Network

Input-output linkages amplify shocks across sectors.

Leontief inverse

$$(I - \Omega^H)^{-1} = I + \Omega^H + (\Omega^H)^2 + \cdots$$

Domar weights

$$\lambda' = \beta'(I - \Omega^H)^{-1}$$

Total foreign exposure

$$\mathcal{M} = (I - \Omega^H)^{-1}\omega_F$$

Interpretation

- Domar weights measure total production importance.
- Indirect input linkages amplify shocks.
- Import exposure propagates through the same network.

Foreign Sector

Foreign prices, interest rates, and demand are exogenous.

Law of One Price

$$P_t^{FH} = S_t P_t^{F*}$$

UIP

$$i_t = i^* + E_t \Delta e_{t+1} + \psi(\hat{b}_t^*) + RP_t$$

Net foreign assets

$$\hat{b}_t^* = \frac{1 + i_{t-1}}{\Pi_t^C} \hat{b}_{t-1}^* + \frac{P_t^H EX_t - P_t^{FH} IM_t}{P_t^C}$$

Transmission

- Exchange rate changes import costs.
- Exchange rate changes export demand.
- Both channels affect inflation and output.

Price Setting

Calvo pricing delivers a network Phillips curve.

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})E_t\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma\Delta e_t$$

Network coefficients

$$\mathcal{B} = \frac{\tilde{A}\alpha}{\kappa}, \qquad \Gamma = \frac{\tilde{A}\omega_F}{\kappa}$$

Key insight

- Labor and import costs propagate through the same network.
- Exchange-rate shocks become an additional cost-push channel.
- Only relative TFP shocks affect inflation.

Monetary Policy

The central bank targets the Divine Coincidence (DC) index.

Free float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$$

Hard peg

$$\hat{s}_t = 0$$

Managed float

$$\hat{i}_t = \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \hat{s}_t$$

The policy rule is the only equation that changes across regimes.

Equilibrium

The model is solved as a nonlinear DSGE in Dynare.

Four equilibrium blocks

Phillips Curve

IS Curve

UIP + NFA

Policy Rule

Changing the monetary policy block generates the three exchange-rate regimes while all remaining equations stay unchanged.

Model: Overview

Households

CRRA consumption, disutility of labor; **Euler eq.**: intertemporal consumption smoothing; labor supply: wage = MRS

UIP (derived, not assumed): domestic rate = foreign rate + risk premium + expected depreciation

Firms

Cobb-Douglas: labor + domestic network inputs + imports

Marginal cost moves with A_{it} (TFP) **and** S_t (import cost) — **two cost-push channels**

Calvo pricing: reset prob. δ_i, forward-looking reset price; sector-specific stickiness $\rightarrow$ heterogeneous slopes

Open Economy

Law of one price: import price = $S_t \times$ foreign price; **UIP** closes the FX block

NFA law of motion: current account accumulates foreign bonds; exports: foreign demand $\times$ relative-price elasticity

Key takeaway: same network structure as Rubbo (2024), **plus** an exchange-rate cost-push channel running through imported inputs.

$\rightarrow$ full equations & notation: Appendix

Model: Policy Regimes and Shocks

Regime	Rule	Exchange rate
Float	$i_t = \bar{\imath} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t$	fully endogenous (UIP)
Peg	$S_t = \bar{S}$	fixed; i_t residual
Managed	$i_t = \bar{\imath} + \phi_\pi \pi_t^{DC} + \phi_y \tilde{y}_t + \phi_s \log(S_t/\bar{S})$	partially stabilized

Shocks (AR(1)): a_{it} (TFP, ρ=0.90) p_t^{F*} (import price, ρ=0.85) D_t^* (foreign demand, ρ=0.80)

Theory: Regime Comparison — Predictions

	Float	**Peg**	**Managed**
Exchange rate	endogenous	fixed	partial
FX cost-push absorbed by	S_t	π_i, y	split
Monetary independence	full	none	full

Naive: float best. **Overturned** — see Welfare section.

Does the DC Index Survive a Second Cost-Push Channel?

Closed economy (Rubbo): 1 channel (TFP) $\Rightarrow$ unique DC index

$$\boldsymbol{\phi}^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \textstyle\sum_i w_i^{DC} \pi_{it}$$

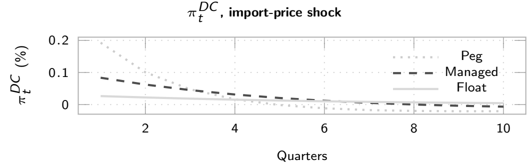
Open economy: 2 channels (TFP + FX) $\Rightarrow$ need *both*, same weights ϕ

$$\underbrace{\boldsymbol{\phi}^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\boldsymbol{\phi}^\top \Gamma = 0}_{\text{FX}} \Rightarrow \textbf{generically infeasible}$$

Answer: no, generically

This calibration: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ — close, but not constant

Fix: flexible import pricing ($\hat{\delta}_M$=1) restores π_t^{DC}



Full DC-index algebra, proof, and knife-edge test: Appendix.

Calibration: Chile, Real IO Data

Parameter	Value	Parameter	Value
β	0.99	$\alpha_1, \alpha_2, \alpha_3$	0.503, 0.359, 0.604
γ, φ	1.00, 2.00	$\Omega_1^F, \Omega_2^F, \Omega_3^F$	0.077, 0.195, 0.070
η	1.50	$\beta_1^H, \beta_2^H, \beta_3^H$	0.027, 0.229, 0.744
ε	8.00	$\delta_1, \delta_2, \delta_3$	0.90, 0.31, 0.16
θ^*	2.00	ψ	0.020
$\beta^{F,tot}$	0.10	ϕ_π, ϕ_y, ϕ_s	1.50, 0.50, 0.30

Dense Ω^H (order: Res., Man., Serv.), all 9 entries used, no triangular assumption:

$$\Omega^H = \begin{pmatrix} 0.075 & 0.153 & 0.193 \\ 0.099 & 0.202 & 0.145 \\ 0.002 & 0.058 & 0.266 \end{pmatrix}, \quad \text{Export share} = (0.602,\ 0.180,\ 0.036)$$

Resource = mining/copper (Chile's export engine, 60% of own output exported) $\rightarrow$ Manuf. (mid-chain) $\rightarrow$ Services (largest value added, sticky)[2]

$\alpha_i, \Omega_{ij}^H, \Omega_i^F, \beta_i^H$: **data** (Banco Central de Chile, Cuadros 12x12, CdeR 2018, ref. 2023)

$\delta_1, \delta_2, \delta_3$: **literature** — euro-area price-change frequencies (Dhyne et al. 2005/ECB IPN) $\rightarrow$ quarterly reset prob.; cross-checked vs. Nakamura & Steinsson (2008) US PPI

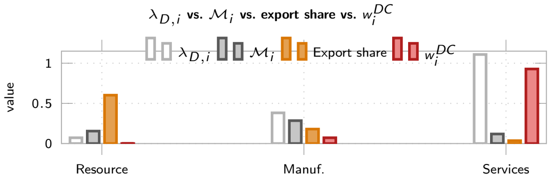
$\beta, \gamma, \varphi, \eta, \varepsilon, \theta^*, \psi, \phi_\pi, \phi_y, \phi_s$: literature-standard. Solved nonlinear, order 1

[2] Sector concordance (Banco Central de Chile, CdeR 2018, 12 activities $\rightarrow$ our 3): **Resource** = {1 Agriculture, forestry & fishing, 2 Mining}; **Manufacturing** = {3 Manufacturing industry, 4 Electricity, gas, water & waste management, 5 Construction}; **Services** = {6 Trade, hotels & restaurants, 7 Transport, communications & information services, 8 Financial intermediation, 9 Real estate & housing services, 10 Business services, 11 Personal services, 12 Public administration}. Import & export data exist at the sector level from the same tables: imports = imported intermediate input use per buying sector (sheet 21, underlies Ω_i^F); exports = exports of each sector's own output (sheet 20, underlies the export-share figures above).

Calibration: Network Properties

$$\lambda_{D,i} = \left[(\boldsymbol{\beta}^H)^\top (I - \Omega^H)^{-1}\right]_i, \quad \mathcal{M}_i = \left[(I - \Omega^H)^{-1}\Omega^F \mathbf{1}\right]_i, \quad w_i^{DC} \propto \lambda_{D,i}\, \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

$\lambda_{D,i}$: Domar weight $\mathcal{M}_i$: import centrality w_i^{DC}: DC weight



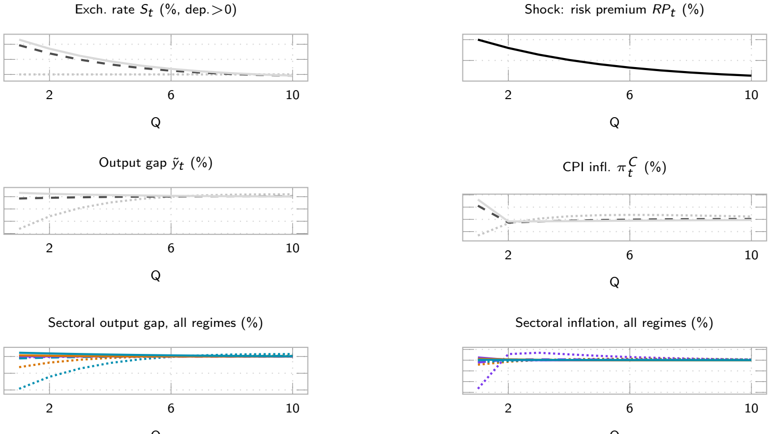
FX exposure runs **opposite** to Domar weight

Resource: small weight (0.07), most FX-exposed ($\mathcal{M}_i = 0.16$) — and *is* the export engine (60% of own output exported, copper)

Services: dominates output ($\lambda_D = 1.11$), FX-exposed only **indirectly**; w_i^{DC} is essentially a Services index (0.93)

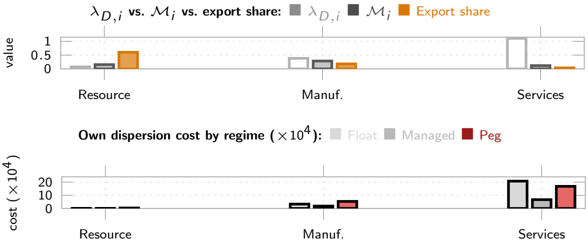
IRFs: Foreign Risk-Premium Shock — the Key Mechanism

The shock that decides the ranking: 75% of Peg's welfare loss (Chile calibration).



Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

Network Exposure & Sector Welfare Preferences



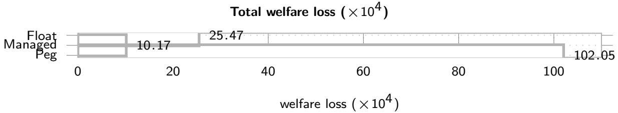
Exposure $\rightarrow$ pass-through $\Gamma_i = \frac{[\Delta(I-\Omega\Delta)^{-1}\boldsymbol{\omega}_F]_i}{\kappa_i}$, **welfare weight** $\kappa_i = \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}$:

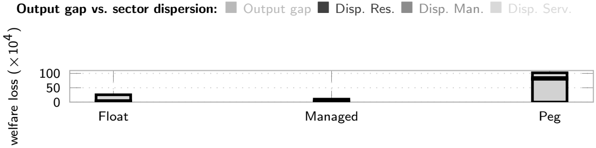
	$\mathcal{M}_i$	Export share	Γ_i	κ_i
Resource	**0.155**	**0.602**	**1.091**	0.07
Manuf.	0.285	0.180	0.001	21.5
Services	0.119	0.036	$<$0.0001	**276.3**

Under literature-sourced Calvo stickiness ($\hat{\delta}_3 = 0.03$), κ_i concentrates in Services — FX pass-through barely matters for Services' welfare cost *relative to its own weight*: almost all of Γ_i funnels through Resource

Managed dominates every sector. Peg costliest for Resource (0.37) & Manuf. (5.36); for Services, **Float is costliest** (20.73 $>$ Peg's 16.79) — near-unit-root S_t under Float feeds the stickiest sector's marginal cost for a long time

Welfare: Headline & Persistence



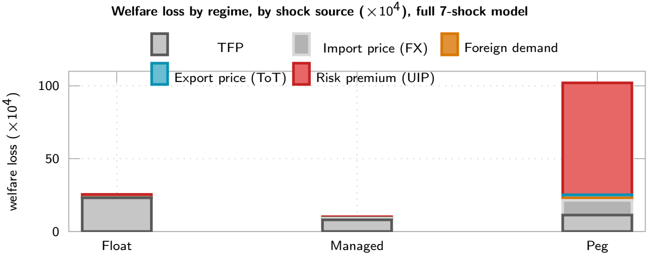


$\mathcal{W} = \frac{1}{2}\left[(\gamma+\varphi)\mathrm{Var}(\tilde{y}_t) + \sum_i \lambda_{D,i}\varepsilon\frac{1-\hat{\delta}_i}{\hat{\delta}_i}\mathrm{Var}(\pi_{it})\right]$ — **Peg dominated** on output gap & dispersion, every sector.

Why: persistence, not volatility.

Float's S_t (near-unit-root NFA) feeds *every* sector's MC, even for domestic shocks
Peg's output-gap term explodes (79.5) because it has zero monetary autonomy to absorb the risk-premium shock
Slow decay, not larger impact, drives Float's Services dispersion above Managed's

Shock Decomposition of Welfare Cost



Risk-premium/UIP dominates Peg (75% of its total)

Ranking: Managed $<$ Float $\ll$ Peg — Peg no longer competitive

2nd-order check: this is the channel a genuine order-2 solution could move most — it does move most, but only $+0.7\%$ *(see Conclusion & appendix).*

Mechanisms: Why Each Regime Costs What It Costs

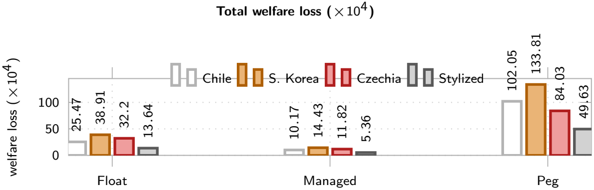
Regime	Cost channel	What sets the magnitude
Float	S_t's near-unit-root persistence (NFA state) feeds $\Gamma_i \Delta e_t$ into *every* sector's MC, even for shocks with no FX content — pure dispersion cost	Persistence of debt-elastic premium (ψ); Γ_i (import exposure); κ_i of the most central/sticky sector (Services: 20.7 of 25.5)
Peg	Zero monetary autonomy: output-gap term explodes (79.5 of 102.1, $\approx$78%) on the UIP shock; $\Delta e_t \equiv 0$ *relocates*, not removes, adjustment into upstream P^H	Absence of ϕ_π, ϕ_y; real vs. nominal share of the shock; upstream sector's Domar weight
Managed	$\phi_S \Delta \log S_t$ damps $\Gamma_i \Delta e_t$ without fully re-imposing peg's relocation cost — dominates every sector at once	Interior optimal $\phi_S^* \approx 0.20$: too low $\rightarrow$ UIP shock leaks in; too high $\rightarrow$ autonomy lost

Two forces, always in tension: Γ-channel favors peg (switches off $\Gamma_i \Delta e_t$); **relocation** channel favors float (peg's fixed e_t forces the same adjustment into upstream prices). Which wins depends on Γ_i vs. κ_i; managed is the only regime improving on *both* corners for every sector.

Analytical or simulated? Both.

Channels — closed-form: $\Gamma_i = [\tilde{A}\boldsymbol{\omega}_F]_i/\kappa$ (pass-through), $\kappa_i = \lambda_{D,i}\varepsilon \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$ (welfare weight), DC-survival $\boldsymbol{\phi}^\top \mathcal{V}=\mathbf{0}^\top$ *and* $\boldsymbol{\phi}^\top \Gamma=0$ — from primitives alone, no model solve.
Magnitudes/ranking — need simulation: $\mathrm{Var}(\pi_{it}), \mathrm{Var}(\tilde{y}_t)$ come from the full linear RE solution (order-1 Dynare) — no closed form once $\phi_S>0$ or Ω^H dense.
Managed$<$Float$\ll$Peg and $\phi_S^*\approx$0.20 are simulated, but robust across all four calibrations.

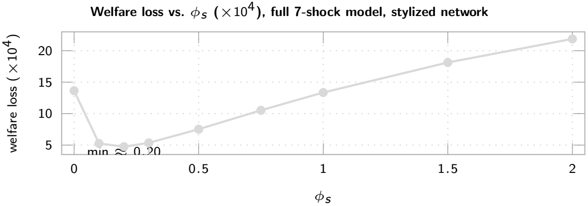
Robustness: South Korea, Czechia & the Stylized Baseline



South Korea: diversified manufacturing exporter (Bank of Korea, 2022), deeper manufacturing import intensity than Chile ($\Omega^F_{\text{manuf}} = 0.24$ vs. 0.19)
Czechia: third real IO calibration (Eurostat, 2022), deepest manufacturing import intensity ($\Omega^F_{\text{manuf}} = 0.32$, EU/German supply chains)
Stylized: original hand-picked triangular Ω^H (Rubbo-style, no data) — checks the result isn't an artifact of network shape
Ranking survives in all four: Managed $<$ Float $\ll$ Peg

UIP dominates Peg's loss in all four: 75% Chile, 72% Korea, 65% Czechia, 81% stylized — not specific to one calibration. All solved nonlinear (Dynare), not linearized.

Generalization: Optimal FX Stabilization ϕ_s

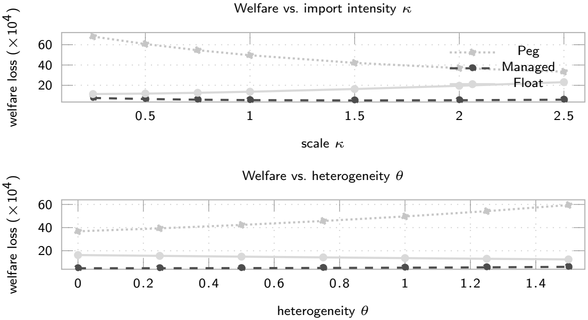


Why interior: too low → UIP shock leaks in; too high → loses autonomy

ϕ_s=0 → 0.20: loss 13.64 → 4.74. **Peaked, not flat:** 0.20 beats calibrated 0.30 by ≈13%

Beyond ≈ 0.2: rises fast, back above Float by ϕ_s=1

Generalization: Robustness



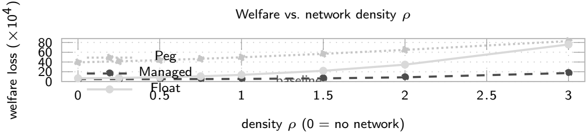
Managed $<$ **Float** $<$ **Peg robust** (stylized network, as in the FX-stabilization sweep), but the gap moves in **opposite directions**:

κ (import scale): Peg falls, Float rises — **converge**

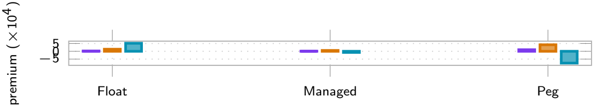
θ (concentration): Peg rises, Float falls — **diverge**

Isolating the Network Channel

New sweep, stylized triangular network (not Chile's dense Ω^H): scale domestic I-O links OH_{21}, OH_{32} by ρ. ρ=0: no domestic network. ρ=1: baseline.



Network welfare premium by sector ($\times 10^4$), total(ρ=1)−total(ρ=0): Resource Manuf. Services



Managed is almost network-neutral: ρ=0 → 1 adds only +0.35 to Managed vs. +6.98 Float, +10.30 Peg
Denser network ⇒ corners diverge from Managed: by ρ=3, Float (75.9) catches Peg (82.9), both ≫ Managed (17.4)
Burden lands on different sectors: Peg/Managed hit upstream Resource/Manuf., Services *gains*; Float hits Services most
Verdict: network **triples** the regime gap as density rises (ρ:1 → 3) and **redirects** who bears the cost

Conclusion

Managed float optimal: 10.17 vs. Float 25.47, Peg 102.05 ($\times 10^4$, Chile calibration)

Driver: risk-premium/UIP dominates Peg (75%); TFP persistence dominates Float & Managed

Optimal $\phi_s \approx 0.20$: sharply peaked — over-stabilizing S_t is costly

Ranking robust to κ, θ, ρ (network density) & to Korea/Czechia calibrations; the *gap* is not

Network channel isolated (stylized network, ρ dial): ρ=0 $\rightarrow$ 1 adds only +0.35 to Managed vs. +6.98 Float, +10.30 Peg — **denser networks favor Managed most**, and reallocate the network's sectoral burden (Services gains under Peg, loses under Float)

Vs. literature: same ranking as Galí–Monacelli, different channel (risk-premium, not ToT)

2nd-order-verified (Dynare order=2, pruned, simulated): loss rises $\leq$ 2% net of MC noise, ranking unchanged — see appendix

Next: many-sector OECD TiVA calibration (beyond the 3-sector Chile/Korea/Czechia data)

Appendix: Is the Ranking Robust to a Genuine 2nd-Order Solution?

Why this was a caveat: Rubbo's LQ welfare formula is 2nd-order in utility, but evaluated on *1st-order* (certainty-equivalent) paths — valid only if $\mathbb{E}[\tilde{y}_t] = \mathbb{E}[\pi_{it}] = 0$ exactly. Two open-economy features can break that: the **debt-elastic UIP wedge** (nonlinear in B_t^*) and the **near-unit-root NFA** process — both mean the true *risk-adjusted* steady state can sit away from the deterministic one.

Method: the model is already coded nonlinear (levels) in Dynare. Solve to **order 2 with pruning**, **simulate** 260k periods (analytic order-$\geq$ 2 moments are unavailable: the Taylor rule leaves price levels/S with a unit root). Compute $\mathbb{E}[X^2]$ directly (not Var) so any risk-adjusted mean shift is captured. Control for simulation noise with an identical-seed order=1 run.

Welfare loss ($\times 10^4$)	**Float**	**Managed**	**Peg**
1st-order (analytic, headline)	25.47	10.17	102.05
1st-order (simulated, same seed)	25.58	10.20	102.39
2nd-order (pruned, simulated)	**26.09**	**10.26**	**103.13**
Net 2nd-order effect	+2.0%	+0.6%	+0.7%

Ranking survives: Managed < Float < Peg at 2nd order, correction $\leq$ 2% everywhere

Appendix: DC Index — Definition and Insulation

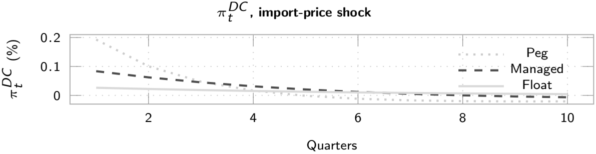
Closed economy (Rubbo): 1 channel (TFP), $\mathcal{V}$ rank-deficient $\Rightarrow$ unique DC index:

$$\boldsymbol{\phi}^\top \mathcal{V} = \mathbf{0}^\top \Rightarrow \pi_t^{DC} = \textstyle\sum_i w_i^{DC} \pi_{it}, \; w_i^{DC} \propto \lambda_{D,i} \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$$

Open economy: 2 channels (TFP + FX):

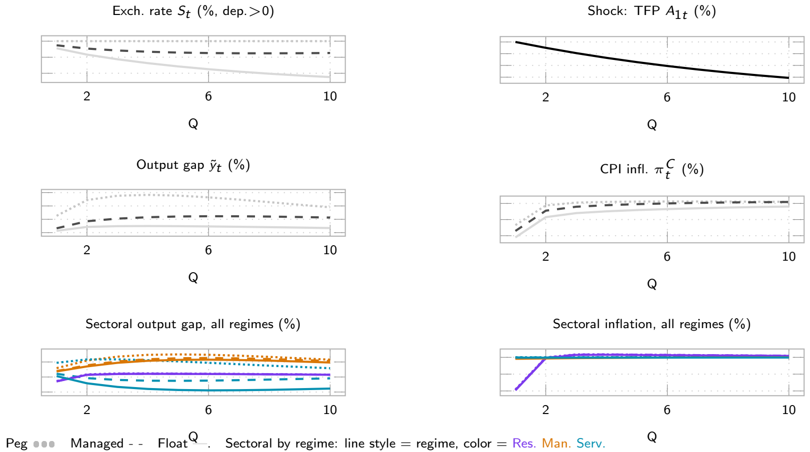
$$\underbrace{\boldsymbol{\phi}^\top \mathcal{V} = \mathbf{0}^\top}_{\text{TFP}} \text{ and } \underbrace{\boldsymbol{\phi}^\top \Gamma = 0}_{\text{FX}} \Rightarrow \textbf{generically infeasible}$$

Knife-edge test: $\Gamma \propto \mathcal{B}$ would save the DC index. Instead: $\cos(\Gamma, \mathcal{B}) \approx 0.96$ ($\approx 17^\circ$ apart) — close, but $\Gamma_i/\mathcal{B}_i$ shrinks $\approx$2$\times$ per sector, not constant. Breakdown is real, not a numerical artifact.
Fix: import nodes flexible ($\hat{\delta}_M$=1 $\Rightarrow$ w_M=0) $\Rightarrow$ DC well-defined again by construction — what the Taylor rules target throughout (π_t^{DC}); FX insulation itself is a quantitative question, shown below.

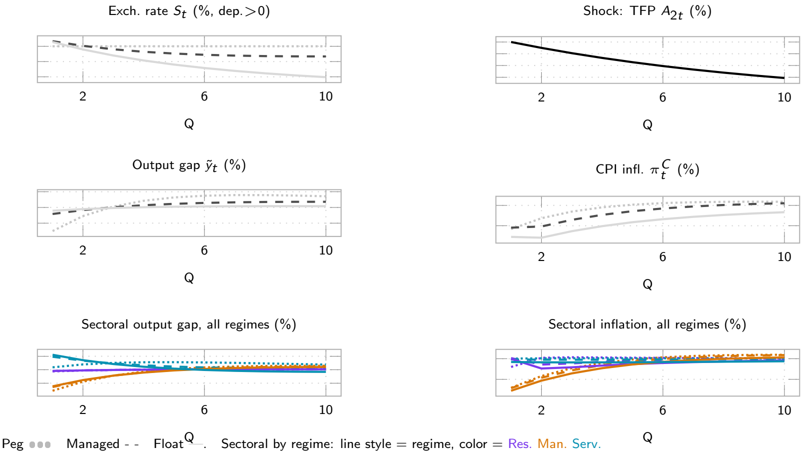


Float: smallest on impact (0.026%) & most persistent; Peg/Managed overshoot and reverse sign — on-impact insulation $\neq$ full story.

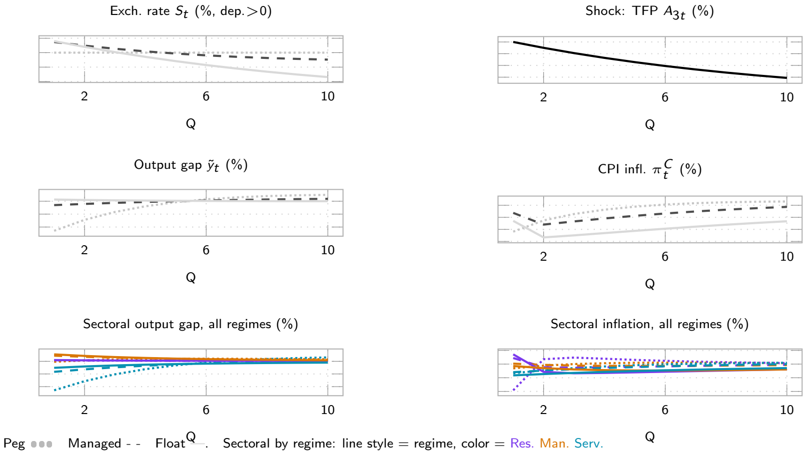
IRFs: Resource TFP Shock



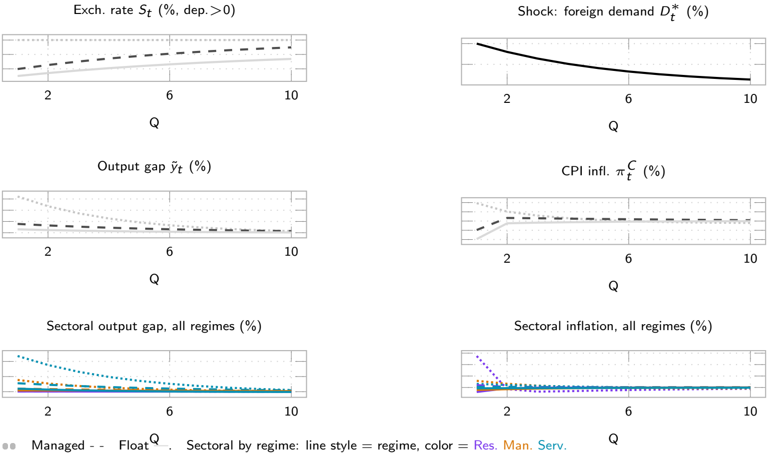
IRFs: Manufacturing TFP Shock



IRFs: Services TFP Shock

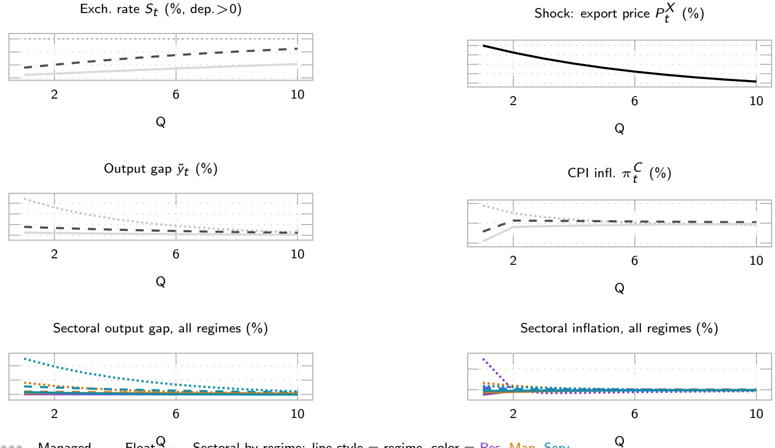


IRFs: Foreign-Demand Shock



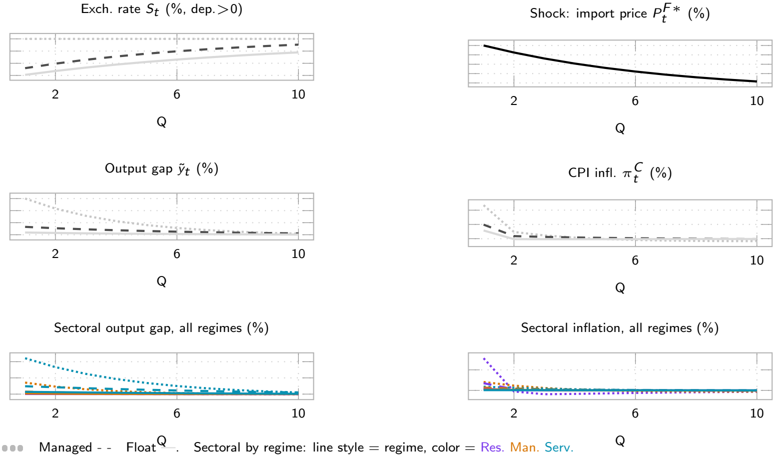
Peg ••• Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

IRFs: Export-Price / ToT Shock



Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

IRFs: Import-Price Shock (Backup)



Peg ●●● Managed - - Float —. Sectoral by regime: line style = regime, color = Res. Man. Serv.

Appendix: Model Equations & Notation (1/2)

Households

$$\max\ \mathbb{E}_0 \sum_t \beta^t \Big[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \Big] \quad \text{s.t.} \quad P_t^C C_t + S_t B_t^* = W_t L_t + \Pi_t + (1+i^*)[1-\psi(\hat{b}_{t-1}^* - \bar{b}^*)] S_t B_{t-1}^*$$

$$C_t^{-\gamma} = \beta(1+i_t)\mathbb{E}_t[C_{t+1}^{-\gamma}/\Pi_{t+1}^C], \quad W_t/P_t^C = C_t^\gamma L_t^\varphi, \quad (1+i_t) = (1+i^*)[1-\psi(\hat{b}_t^* - \bar{b}^*)]\, \mathbb{E}_t[S_{t+1}/S_t] \text{ (UIP)}$$

C, L: consumption, labor — β, γ, φ: discount, RRA, inv. Frisch — W, i: wage, dom. rate — S, i^*: ER, foreign rate
ψ: debt-elastic premium — B^*: foreign bond — P^C: CPI — Π_t: profits (rebated)

Firms: Technology and Cost

$$Y_{it} = A_{it} L_{it}^{\alpha_i} \prod_j X_{ijt}^{\Omega_{ij}^H} M_{it}^{\Omega_i^F} \quad \Rightarrow \quad MC_{it} = \frac{W_t^{\alpha_i} \prod_j P_{jt}^{\Omega_{ij}^H} (S_t P_t^{F*})^{\Omega_i^F}}{A_{it}}$$

$$W_t L_{it} = \alpha_i MC_{it} Y_{it}, \quad P_{jt} X_{ijt} = \Omega_{ij}^H MC_{it} Y_{it}, \quad S_t P_t^{F*} M_{it} = \Omega_i^F MC_{it} Y_{it}$$

Y, A: output, TFP — L, X, M: labor, dom./imp. inputs — Ω_{ij}^H: dom. IO share — Ω_i^F: import cost share
α_i: labor share — MC_{it}: nominal marg. cost — P^{F*}: foreign import price

Key: MC_{it} moves with both A_{it} (TFP) and S_t (via M_{it}) — the two cost-push channels.

Appendix: Model Equations & Notation (2/2)

Firms: Calvo Pricing — reset price (probability δ_i resets, $1-\delta_i$ keeps last price):

$$X1_{it} = \widetilde{MC}_{it} Y_{it} + (1-\delta_i)\beta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma} \Pi^{\varepsilon}_{i,t+1} X1_{i,t+1}, \quad X2_{it} = Y_{it} + (1-\delta_i)\beta\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma} \Pi^{\varepsilon-1}_{i,t+1} X2_{i,t+1}$$

$$P^{\#}_{it}/P_{it} = \tfrac{\varepsilon}{\varepsilon-1}\, X1_{it}/X2_{it}, \quad 1 = (1-\delta_i)\Pi^{\varepsilon-1}_{it} + \delta_i (P^{\#}_{it}/P_{it})^{1-\varepsilon}, \quad \hat{\delta}_i = \frac{\delta_i(1-\beta(1-\delta_i))}{1-\beta\delta_i(1-\delta_i)}$$

δ_i: Calvo reset prob. (primitive) $\quad$ $\hat{\delta}_i$: derived slope (enters DC weights) $\quad$ $P^{\#}_{it}$: reset price $\quad$ ε: elasticity (markup $\varepsilon/(\varepsilon-1)$)

Open Economy

$$\underbrace{P^{FH}_t = S_t P^{F*}_t}_{\text{LOP}}, \quad \underbrace{(1+i_t) = (1+i^*)\left[1-\psi(\hat{b}^*_t - \bar{b}^*)\right] \mathbb{E}_t[S_{t+1}/S_t]}_{\text{UIP}}$$

$$\hat{b}^*_t = \frac{1+i_{t-1}}{\Pi^C_t}\hat{b}^*_{t-1} + \frac{P^H_t EX_t - P^{FH}_t IM_t}{P^C_t}, \quad EX_t = \kappa_{EX} D^*_t \left(\frac{P^H_t}{P^{FH}_t}\right)^{-\theta^*}$$

$$Y_{it} = \beta^H_i \frac{P^H_t}{P_{it}}(C^H_t + EX_t) + \sum_j \Omega^H_{ji}\frac{MC_{jt}Y_{jt}}{P_{it}}$$

$\hat{b}^*$: NFA (scaled) $\quad$ EX, IM: exports, imports $\quad$ θ^*: export elasticity $\quad$ D^*: foreign demand shifter

β^H_i: cons. share, sector i $\quad$ Float: i_t Taylor rule, S_t free $\quad$ Peg: $S_t = \bar{S}$, i_t residual

Appendix: The Peg vs. Float Debate — What Do We Know?

Why it depends (structure):

- **Mundell (1961)** OCA — fix favored by small, open, factor-mobile economies w/ correlated shocks vs. anchor; float favored by large, diversified ones
- **Frankel (1999)** — "no single regime is right for all countries or at all times"; openness, size, trade & capital-market structure all matter

Shock-absorption view (why float wins on real shocks — robustly):

- **Friedman (1953)** — S_t adjusts instead of costly wage/price deflation
- **Broda (2004)** — 75 dev. countries, 1973–96: negative ToT shocks $\rightarrow$ large real GDP declines under pegs
- **Edwards & Levy-Yeyati (2005)** — 10% ToT deterioration: -0.80pp growth under pegs vs. -0.43pp under floats
- **Céspedes, Chang & Velasco (2004)** — even with dollarized-liability balance-sheet effects, float *still* beats fixed

But floating is not a free lunch (limits on float's insulation):

- **Calvo & Reinhart (2002)** "Fear of Floating" — EMEs peg de facto anyway: credibility-building, not literal welfare optimum
- **Rey (2013)** "Dilemma not trilemma" — under free capital mobility, float alone doesn't deliver monetary autonomy

Consensus: no corner solution is universally optimal — the choice trades a nominal anchor against real-shock absorption, shock- and structure-dependent.

Appendix: The Frontier — Is Fear of Floating Actually Optimal?

Itskhoki & Mukhin (2023) — general open-economy framework with endogenous UIP/PPP deviations: efficient policy = MP closes the output gap + FX intervention eliminates the UIP wedge. Optimal policy is generically a **managed float/crawling peg**, matching observed fear of floating. When the natural RER is stable, this collapses to an "open-economy divine coincidence."
$\hookrightarrow$ *vs. us:* **same headline** (Managed, not a corner, is optimal) — reached instead via a multi-sector Calvo network with a persistence-driven, not UIP-gap-closing, mechanism

Bianchi & Coulibaly (2025; 2022 WP) — fear of floating is **optimal**, not behavioral, when credit access is tied to the real exchange rate (self-fulfilling crisis risk). Prescription: float for real shocks, **fix** for financial shocks
$\hookrightarrow$ *vs. us:* **opposite ranking** for our financial shock — our risk-premium/UIP shock passes straight through under **Peg** (worst regime). Different financial friction (persistent NFA wedge vs. collateral-constraint crisis) flips the prescription

Coulibaly (2023) — discretionary MP is procyclical to offset balance-sheet effects; commitment (inflation targeting) or capital controls raise welfare
$\hookrightarrow$ *vs. us:* echoes our "Managed = extra instrument" logic — giving policy more tools beats a corner regime

Bottom line: the 2023–25 frontier has also moved past corner solutions — but *why* fear of floating is optimal is mechanism-specific. We add production-network amplification + shock persistence as a distinct reason, not examined by either paper.

Appendix: How Our Results Compare to the Literature

Literature	Says	Us
Mundell (1961); Frankel (1999)	Optimal regime depends on structure, no universal answer	**Confirms**: ranking flips with shock type (UIP vs. TFP) and network density
Friedman (1953); Broda (2004); Edwards–Levy-Yeyati (2005); Céspedes–Chang–Velasco (2004)	Float buffers real (ToT/foreign) shocks, lower output loss than peg — robust even to balance-sheet effects	**Same direction** (Peg worst), **different shock**: risk-premium/UIP, not ToT
Galí & Monacelli (2005)	Peg dominated, float $\approx$ optimal in 1-sector SOE	**Same ranking of Peg**, but pure **Float is not optimal**: Managed beats it
Calvo–Reinhart (2002)	EMEs avoid pure float — read as **fear**, a credibility-driven deviation from optimum	We find partial stabilization is **literally welfare-optimal** — a persistence-driven, not fear-driven, reason
Rey (2013)	Float alone doesn't insulate; need a second instrument (capital controls/macropru)	Echoes Fanelli–Straub: Managed uses S_t itself as that **second instrument**
Qiu, Wang, Xu & Zanetti (2026, JME)	Open-economy DC index (CPI+expenditure-switching+profit channels), *static*, WIOD-calibrated; DC breaks down, output-gap targeting near-optimal	**Independent corroboration** DC breaks down in open economy; **we add** the piece they flag as their own top open extension — UIP, NFA persistence, and FX-**regime choice**

Appendix: The ECB's Open-Economy Production-Network Model

Gnocato, Montes-Galdón & Stamato (2025), "Tariffs across the supply chain," ECB WP No. 3081 — also underlies Lane's (ECB Exec. Board) 13 May 2026 speech on energy shocks (4-region extension):

- **Setup:** two-region, multi-sector, full IO structure, international production network; sticky wages/prices; Taylor rule on *domestic* CPI (no regime comparison)
- **Finding:** tariffs on **intermediate** inputs → persistent inflation & larger GDP loss (low substitutability, network propagation); tariffs on **final** goods → one-off inflation spike, smaller GDP loss — *where* in the network a shock hits sets its persistence
- ↪ *vs. us:* their tariff-location result and our TFP-vs-UIP result are the same insight in two settings: shock **location/type** sets its persistence, echoing our "driver is persistence, not volatility"

What it cites for open economy + production networks:

- Network amplification, closed economy: **Pasten, Schoenle & Weber (2020)**; **Ghassibe (2021)**; **Rubbo (2023)**[3]; **Afrouzi & Bhattarai (2023)**
- Open-economy multi-sector NK + networks: **Devereux, Gente & Yu (2023)**; **Hinterlang et al. (2023)** EMuSe (Bundesbank); closest: **Kalemli-Özcan, Soylu & Yildirim (2025)** "Global networks, MP and trade"
- Trade fragmentation & inflation: **Moro & Nispi Landi (2024)**; **Ambrosino, Chan & Tenreyro (2026, JIE)**
- Optimal MP under tariffs: **Bergin & Corsetti (2023)**; **Bianchi & Coulibaly (2025)** "Optimal MP response to tariffs" (*different* paper from their fear-of-floating one)

Bottom line: none compares FX **regimes** or asks whether a DC index survives — they fix the ~~monetary rule. That gap is still exactly~~ our question.

[3] Published version: Rubbo, E. (2023), Econometrica 91(4), 1417–1455 — journal year is 2023, not 2024.